

# CariSurg Healthcare AI Program 2026 - Week 0

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## *Assignment 6 - Triage Pseudocode (At-Risk Patient Logic)*

Building on the DBP cleaning from Assignment 2 and the visualisations from Assignment 3, I designed a rule-based algorithm that checks each patient's vital signs against clinically validated normal ranges. If any single vital sign falls outside its threshold the patient is flagged as at risk.

Triage Pseudocode:

for each patient in the dataset:

if pulse < 60 or pulse > 100 bpm:

flag as AT RISK

# below 60 indicates bradycardia, above 100 indicates tachycardia

# both can signal cardiovascular instability

if temperature < 36.0 or temperature > 38.0 degrees C:

flag as AT RISK

# below 36 indicates hypothermia, above 38 indicates fever or infection

if respiratory rate < 12 or respiratory rate > 20 breaths per minute:

flag as AT RISK

# outside this range can indicate respiratory distress or failure

if systolic blood pressure < 90 or systolic blood pressure > 140 mmHg:

flag as AT RISK

# below 90 indicates hypotension or possible shock

# above 140 indicates hypertension which increases stroke risk

if DBP < 60 or DBP > 90 mmHg:

flag as AT RISK

# below 60 indicates hypotension risk

# above 90 indicates hypertension concern

otherwise:

flag as NOT AT RISK

Note: Values checked against WHO Emergency Triage guidelines.

The only limitation I can see is the anomalies to this rule such as athletes who would have a naturally low resting rate and either high or low baselines to these metrics.